Niloofar Razi

Qazvin, Iran +98 (910) 4693066
niloofar.razi98@gmail.com Niloofar-Razi

RESEARCH INTERESTS

- AI/ML for multi-objective optimization in construction
- Data-driven scheduling and resource allocation under uncertainty
- XR and digital twins for planning and training
- Sustainable construction (energy use and CO_2)
- Safety analytics and risk modeling

EDUCATION

September 2020 – July 2023
- **M.Sc**. in Construction Engineering and Management
- Imam Khomeini International University (IKIU), Qazvin, Iran.
- **Thesis**: "Time, Cost, Energy Consumption, and CO_2 emissions Trade-off Model Optimization in construction projects using Machine Learning Algorithms"
- **Supervisor**: Dr. Ramin Ansari
- **GPA:** 19.47 out of 20

September 2016 - August 2020
- **B.Sc**. in Civil Engineering
- Imam Khomeini International University (IKIU), Qazvin, Iran.
- **GPA:** 16.59 out of 20

PUBLICATIONS

Published Journal Article

Razi, N., & Ansari, R. (2024). *A prediction-based model to optimize construction programs: Considering time, cost, energy consumption, and CO_2 emissions trade-off.* ***Journal of Cleaner Production****, 445, 141164. https://doi.org/10.1016/j.jclepro.2024.141164*

Published Journal Article

Razi, N., Badhan, S. J., & Samsami, R. (2026). Artificial Intelligence (AI) in Construction Management (CM): A Systematic Review of Models and Methods. Buildings, 16(11), 2225. https://doi.org/10.3390/buildings16112225

Conference Paper (IEEE)

Razi, N., & Samsami, R. (2025). *Extended Reality (XR) Applications in Civil Engineering Education: A Review*. In Proceedings of the 2025 IEEE 4th International Conference on Intelligent Reality (ICIR) (pp. 1–7). IEEE. https://doi.org/10.1109/ICIR68135.2025.11361605

Manuscript in Preparation

Razi, N. (2025). *Challenges, Solutions, and Future Trends of Artificial Intelligence in Construction Management: A Systematic Review (2020–2025).*

August 2022

Ansari, R., Mirzahossein, H., & **Razi, N.** (2022). Architecture and perspective of deep learning application in construction industry projects. *2nd International Conference on Architecture, Civil Engineering, Urban Development, Environment and Horizons of Islamic Art.*

HONORS AND AWARDS

2023	**Ranked 1st** among Construction Management M.Sc. students
2020	**"Merit-based admission"** (Admission without an entrance exam for M.Sc.)
2020	**Ranked top 5%** among Civil Engineering B.Sc. students
2016	**Ranked 602nd** among more than 156,000 students in the B.Sc. National university entrance exam

SKILLS

Programming Language	• Python		
Data Analysis Tools/Packages	• Scikit-learn • NumPy	• Pandas • TensorFlow	• SciPy
Visualization	• Matplotlib	• Seaborn	• Plotly
Engineering	• ETABS	• AutoCAD	• Microsoft Project (MSP)

LANGUAGE SKILLS

English	• Professional working proficiency (PTE Academic: **Overall 90/90**)
Turkish	• Fluent in Turkish (Istanbul and Azerbaijani dialects)
Persian	• Native proficiency

WORK EXPERIENCE

Oct 2025 – Present	• Web Application Developer, Civil company, Sarisazan Alborz, Qazvin, Iran • Developed a web-based construction project management system featuring dashboards, task tracking, reporting tools, and an AI-assisted workflow interface.
Oct 2023 – Oct 2024	• Technical Office Expert in Civil company, Sarisazan Alborz, Qazvin, Iran
2020 – Apr 2021	• Technical Office Expert in Civil company, Tahkim Emarat Caspian, Qazvin, Iran
2016 – 2020	• English Instructor, Kish_e_Mehr, Qazvin, Iran.

REFERENCES

- **Dr. Reihaneh (Rei) Samsami**
Assistant Professor, University of New Haven, USA (RSamsami@newhaven.edu)

- **Dr. Ramin Ansari**
Faculty Member, Imam Khomeini International University, Iran (raminansari@eng.ikiu.ac.ir)

- **Dr. Mahdi Mahdikhani**
Faculty Member, Imam Khomeini International University, Iran (mahdikhani@eng.ikiu.ac.ir)